Lesson 36: Repeated Games and Folk Theorems

AI/ML Mathematics

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1 Motivation

In one-shot games, players interact once and the outcome is determined by the simultaneous choice of strategies. The Prisoner's Dilemma, for instance, has a unique Nash equilibrium at mutual defection—even though both players would prefer mutual cooperation. This pessimistic conclusion changes dramatically when the game is played *repeatedly*.

Repeated interaction introduces two mechanisms absent from one-shot play:

(i) **History-dependent strategies.** Players can condition current actions on the entire history of past play, enabling reward and punishment schemes.

(ii) **Intertemporal incentives.** A short-term gain from defection can be outweighed by the long-term cost of triggering punishment.

The central results of this lesson—the *folk theorems*—characterize precisely which payoff vectors are sustainable as equilibria in infinitely repeated games. Informally, any feasible payoff that gives each player more than their worst-case guarantee can be achieved, provided players are sufficiently patient.

MARL Training as a Repeated Game

In multi-agent reinforcement learning, agents interact over many episodes and update their policies based on observed outcomes. This is exactly the structure of a repeated game. The folk theorems explain why independent learners can converge to cooperative outcomes (or to complex punishment schemes) that would never arise in single-episode analysis. Understanding these results is essential for predicting and shaping emergent behavior in self-play training.

Roadmap.

- Section 2: Finitely repeated games and the unraveling theorem.
- Section 3: New equilibria when the stage game has multiple Nash equilibria.
- Section 4: Infinitely repeated games with discounting.
- Section 5: Feasible and individually rational payoffs.
- Section 6: Cooperation in the repeated Prisoner's Dilemma via trigger strategies.
- Section 7: The tit-for-tat strategy.
- Section 8: The Nash folk theorem with full proof.
- Section 9: The subgame perfect folk theorem (Fudenberg–Maskin) with full proof.
- Section 10: Connections to reinforcement learning.
- Section 11: Exercises.

2 Finitely Repeated Games

We begin with a *stage game* $G = (N, (A_i)_{i\in N}, (g_i)_{i\in N})$, where $N = \{1, \ldots, n\}$ is the player set, A_i is the finite action set of player i, and $g_i : A \to \mathbb{R}$ is the stage-game payoff function, with $A = \prod_{i\in N} A_i$.

Definition 2.1 (Finitely Repeated Game)**.** The *T-fold repetition* of G, denoted G^T, is defined as follows:

(a) In each period $t \in \{1, 2, \ldots, T\}$, all players simultaneously choose actions $a_i^t \in A_i$.

(b) At the end of period t, all players observe the realized action profile $a^t = (a_1^t, \ldots, a_n^t)$.

(c) The *history* at period t is $h^t = (a^1, a^2, \ldots, a^{t-1}) \in A^{t-1}$. The initial history is $h^1 = \emptyset$.

(d) A *(pure) strategy* for player i is a sequence of functions $s_i = (s_i^1, s_i^2, \ldots, s_i^T)$ where $s_i^t : A^{t-1} \to A_i$ maps histories to actions.

(e) The payoff to player i under strategy profile $s = (s_1, \ldots, s_n)$ is

$$u_i(s) = \frac{1}{T} \sum_{t=1}^{T} g_i\big(a^t(s, h^t)\big),$$

where $a^t(s, h^t) = \big(s_1^t(h^t), \ldots, s_n^t(h^t)\big)$ and histories are generated by the strategy profile.

Remark 2.2. We use average payoffs $\frac{1}{T}\sum_t g_i(a^t)$ rather than total payoffs to keep repeated-game payoffs on the same scale as stage-game payoffs, facilitating comparison with the infinite-horizon discounted case.

Definition 2.3 (SPE for Repeated Games)**.** A strategy profile s^* in G^T is a *subgame perfect equilibrium* (SPE) if, for every period $t \in \{1, \ldots, T\}$ and every history $h^t \in A^{t-1}$, the continuation strategy profile $(s_i^{*,\tau}(h^t, \cdot))_{\tau \geq t}$ is a Nash equilibrium of the subgame starting from history h^t.

2.1 The Unraveling Theorem

The fundamental result for finitely repeated games with a unique stage-game equilibrium is that repetition adds nothing: the unique SPE plays the stage-game NE in every period.

Unraveling Theorem

Theorem 2.4 (Unique Stage-Game NE Implies Unique SPE)**.** *If the stage game G has a unique Nash equilibrium $a^* = (a_1^*, \ldots, a_n^*)$, then the T-fold repeated game G^T has a unique subgame perfect equilibrium in which $a^t = a^*$ for all $t \in \{1, \ldots, T\}$.*

Proof. We proceed by backward induction on t.

Base case ($t = T$). In the final period, regardless of the history h^T, the subgame consists of a single simultaneous play of the stage game G. Since a^* is the unique NE of G, every SPE must prescribe a^* in period T after every history.

Inductive step. Suppose that in every SPE, $a^\tau = a^*$ for all $\tau \in \{t+1, \ldots, T\}$, regardless of the history at period $t+1$. Consider the subgame beginning at an arbitrary history h^t. The continuation payoffs from period $t+1$ onward are fixed at

$$\frac{1}{T} \sum_{\tau=t+1}^{T} g_i(a^*) = \frac{T-t}{T}\, g_i(a^*),$$

which is independent of the action chosen at period t, since all future play is a^* regardless of history. Therefore, player i's total payoff as a function of the period-t action profile a^t is

$$\frac{1}{T}\, g_i(a^t) + \frac{T-t}{T}\, g_i(a^*).$$

Maximizing this over a_i^t is equivalent to maximizing $g_i(a_i^t, a_{-i}^t)$ in the stage game. Since a^* is the unique NE of G, the unique best response profile at period t is $a^t = a^*$ in every SPE.

By induction, $a^t = a^*$ for all $t \in \{1, \ldots, T\}$. Since the action in every period after every history is uniquely determined, the SPE is unique. □

The Unraveling Trap

Theorem 2.4 shows that finite repetition of the Prisoner's Dilemma—which has a unique NE at mutual defection—cannot sustain cooperation under subgame perfection. The "last period" argument propagates backward through every period. This motivates the study of infinitely repeated games, where no last period exists.

3 New Equilibria with Multiple Stage-Game NE

When the stage game has multiple Nash equilibria, repetition *can* create new equilibrium outcomes that are not stage-game NE. The key idea is that different continuation equilibria can serve as rewards and punishments.

Proposition 3.1 (Repetition Enables New Equilibria)**.** *If the stage game G has at least two Nash equilibria yielding different payoffs for some player, then the two-fold repeated game G^2 may have SPE outcomes in which first-period play is not a stage-game NE.*

Proof. We give a constructive proof. Let G have two Nash equilibria a^* and $\hat{a}$ with $g_i(a^*) > g_i(\hat{a})$ for all i (i.e., a^* Pareto-dominates $\hat{a}$). Suppose there exists an action profile $\tilde{a} \notin \mathrm{NE}(G)$ with $g_i(\tilde{a}) > g_i(a^*)$ for all i, and suppose the maximum unilateral gain from deviation satisfies

$$\max_{a_i' \in A_i} g_i(a_i', \tilde{a}_{-i}) - g_i(\tilde{a}) < g_i(a^*) - g_i(\hat{a}). \tag{1}$$

Consider the following strategy for each player i in G^2:

- **Period 1:** Play $\tilde{a}_i$.
- **Period 2:** If $a^1 = \tilde{a}$, play a_i^*. Otherwise, play $\hat{a}_i$.

In period 2, the prescribed play is always a stage-game NE (a^* or $\hat{a}$), so the strategy is sequentially rational in every period-2 subgame.

In period 1, if player i deviates to a_i' while others play $\tilde{a}_{-i}$, the payoff change is:

$$\begin{aligned}
&\frac{1}{2}\big[g_i(a_i', \tilde{a}_{-i}) + g_i(\hat{a})\big] - \frac{1}{2}\big[g_i(\tilde{a}) + g_i(a^*)\big] \\
&= \frac{1}{2}\big[g_i(a_i', \tilde{a}_{-i}) - g_i(\tilde{a})\big] - \frac{1}{2}\big[g_i(a^*) - g_i(\hat{a})\big] < 0,
\end{aligned}$$

where the inequality follows from (1). Hence no player has a profitable deviation, and the profile is an SPE in which first-period play $\tilde{a}$ is not a stage-game NE. □

Example 3.2 (Coordination Game)**.** Consider the two-player coordination game with payoffs:

	L	R
T	$3,3$	$0,0$
B	$0,0$	$1,1$

The game has two pure NE: (T, L) with payoffs $(3, 3)$ and (B, R) with payoffs $(1, 1)$. Suppose we want to sustain the non-NE outcome (T, R) with payoffs $(0, 0)$ in the first period. This fails because the "reward" payoff difference $g_i(T, L) - g_i(B, R) = 2$ would need to exceed the deviation

gain, but $g_i(T, R) = 0$ is already the minimum, so no player gains from unilateral deviation at (T, R)... except player 1 prefers B giving 0 anyway. The construction requires profiles where deviation is *tempting*, which depends on the specific game structure.

A more instructive application: suppose we add an action M for each player with payoffs $g(M, M) = (2.5, 2.5)$, $g(M, L) = g(T, M) = (-1, -1)$, and other entries as before. Then (M, M) Pareto-dominates both NE? No, (T, L) gives $(3, 3) > (2.5, 2.5)$. In any case, the key principle is that the gap between the "reward" and "punishment" equilibria creates room for enforcing non-NE play.

4 Infinitely Repeated Games

To circumvent the unraveling argument, we consider infinite repetition with discounting.

Definition 4.1 (Infinitely Repeated Game)**.** Given a stage game G and a discount factor $\delta \in (0, 1)$, the *infinitely repeated game* $G^\infty(\delta)$ is defined by:

(a) In each period $t \in \{1, 2, 3, \ldots\}$, players simultaneously choose actions $a^t \in A$.

(b) The history at period t is $h^t = (a^1, \ldots, a^{t-1}) \in A^{t-1}$, with $h^1 = \emptyset$. A strategy for player i is a function $s_i = (s_i^t)_{t=1}^\infty$ with $s_i^t : A^{t-1} \to A_i$.

(c) Player i's payoff under strategy profile s is the *normalized discounted sum*:

$$u_i(s) = (1 - \delta) \sum_{t=1}^{\infty} \delta^{t-1} g_i(a^t).$$

The normalization factor $(1-\delta)$ ensures that payoffs lie in the same range as stage-game payoffs: if $a^t = a$ for all t, then $u_i = g_i(a)$.

Remark 4.2. Every history h^t begins a subgame that is strategically identical to $G^\infty(\delta)$ (the continuation game has the same structure and the same discount factor). This self-similar structure is central to the analysis of trigger strategies.

5 Feasible and Individually Rational Payoffs

Definition 5.1 (Feasible Payoff Set)**.** The *feasible payoff set* of the stage game G is

$$V = \operatorname{conv}\{g(a) : a \in A\} = \operatorname{conv}\{(g_1(a), \ldots, g_n(a)) : a \in A\} \subset \mathbb{R}^n,$$

the convex hull of all achievable stage-game payoff vectors.

Remark 5.2. In the infinitely repeated game, any $v \in V$ is achievable by cycling through action profiles with appropriate frequencies. If $v = \sum_{k=1}^K \lambda_k\, g(a^{(k)})$ with $\lambda_k \geq 0$ and $\sum_k \lambda_k = 1$, then playing $a^{(k)}$ in a periodic cycle with frequency λ_k yields average payoff v. As $\delta \to 1$, the normalized discounted payoff converges to this time-average.

Definition 5.3 (Minimax Value and Individually Rational Payoffs)**.** The *minimax value* of player i in pure strategies is

$$\underline{v}_i = \min_{a_{-i} \in A_{-i}} \max_{a_i \in A_i} g_i(a_i, a_{-i}).$$

The *mixed-strategy minimax value* is

$$\underline{v}_i^{\text{mix}} = \min_{\alpha_{-i} \in \Delta(A_{-i})} \max_{a_i \in A_i} \mathbb{E}_{\alpha_{-i}}[g_i(a_i, a_{-i})],$$

where $\Delta(A_{-i}) = \prod_{j \neq i} \Delta(A_j)$. A payoff vector $v \in \mathbb{R}^n$ is *individually rational* if $v_i \geq \underline{v}_i^{\text{mix}}$ for all $i \in N$.

The minimax value is the lowest payoff that the opponents can force upon player i, even when i best-responds. No equilibrium can give i less than this amount.

Proposition 5.4 (NE Payoff Lower Bound)**.** *In any Nash equilibrium of $G^\infty(\delta)$, each player i receives payoff $u_i \geq \underline{v}_i^{\text{mix}}$.*

Proof. In any NE σ, player i can deviate to a strategy that, in each period t, plays $\arg\max_{a_i \in A_i} \mathbb{E}_{\sigma^t_{-i}}[g_i(a_i, a^t_{-i})]$, where σ^t_{-i} is the marginal distribution over opponents' period-t actions induced by the equilibrium strategy. This best-response strategy guarantees a stage payoff of at least $\underline{v}_i^{\text{mix}}$ in every period (since i can guarantee this value against *any* opponents' mixed action profile, and the equilibrium marginal is one such profile). Hence the deviation yields total payoff at least $\underline{v}_i^{\text{mix}}$. Since σ is a NE, the equilibrium payoff must be at least as high as any deviation payoff: $u_i(\sigma) \geq \underline{v}_i^{\text{mix}}$. □

Definition 5.5 (Full Dimensionality Condition)**.** The stage game G satisfies the *full dimensionality condition* if the feasible payoff set $V = \text{conv}\{g(a) : a \in A\}$ has dimension n (the number of players), i.e., V has nonempty interior in $\mathbb{R}^n$.

Remark 5.6. The full dimensionality condition is equivalent to requiring that there exist $n+1$ action profiles $a^{(0)}, a^{(1)}, \ldots, a^{(n)} \in A$ such that the vectors $g(a^{(1)}) - g(a^{(0)}), \ldots, g(a^{(n)}) - g(a^{(0)})$ are linearly independent in $\mathbb{R}^n$. This condition is generic (it holds for "almost all" games) but fails in certain symmetric or degenerate games. For two-player games, it simply requires that the feasible payoff set is not contained in a line.

6 Cooperation in the Repeated Prisoner's Dilemma

Example 6.1 (The Prisoner's Dilemma)**.** Consider the standard Prisoner's Dilemma with payoffs:

	C	D
C	$3,3$	$0,4$
D	$4,0$	$1,1$

The unique stage-game NE is (D, D) with payoffs $(1, 1)$. The minimax value for each player is $\underline{v}_i = 1$ (since mutual defection already holds each player to their minimax). The Pareto-optimal outcome is (C, C) with payoffs $(3, 3)$.

Definition 6.2 (Grim Trigger Strategy)**.** Given a target action profile $a^c \in A$, the *grim trigger strategy* for player i is:

$$s_i^t(h^t) = \begin{cases} a_i^c & \text{if } a^\tau = a^c \text{ for all } \tau < t, \\ m_i & \text{otherwise,} \end{cases}$$

where m_i is player i's component of a minimax punishment profile. The strategy cooperates as long as no deviation has ever occurred, and punishes by minimaxing forever after any deviation.

Proposition 6.3 (Grim Trigger Sustains Cooperation)**.** *In the infinitely repeated Prisoner's Dilemma (Example 6.1) with discount factor δ, the grim trigger strategy profile with $a^c = (C, C)$ and punishment (D, D) is a subgame perfect equilibrium if and only if $\delta \geq 1/3$.*

Proof. We verify the one-shot deviation principle: in any subgame, no player benefits from a single-period deviation followed by reversion to the prescribed strategy.

Cooperation subgame (no deviation has occurred). If player i cooperates, the continuation payoff is 3 (cooperation forever). If player i deviates to D in the current period, the payoff is:

$$u_i^{\text{dev}} = (1 - \delta) \cdot 4 + \delta \cdot 1 = 4 - 3\delta,$$

since the opponent cooperates this period (giving i a stage payoff of 4), but grim trigger punishment begins next period, yielding 1 forever. Cooperation is preferred when $3 \geq 4 - 3\delta$, i.e., $\delta \geq 1/3$.

Punishment subgame (a deviation has occurred). Both players play (D, D) forever. Since (D, D) is a stage-game NE, no player can profitably deviate in any single period: deviating to C gives $(1-\delta) \cdot 0 + \delta \cdot 1 = \delta < 1$, while staying with D gives 1.

Since no profitable one-shot deviation exists in any subgame when $\delta \geq 1/3$, the grim trigger profile is an SPE. Conversely, if $\delta < 1/3$, the deviation in the cooperation phase is profitable, so grim trigger is not an NE (and hence not an SPE). □

Discount Factors in RL

The discount factor δ in repeated games is analogous to the discount factor γ in MDPs. If γ is too small (agents are too myopic), cooperative equilibria cannot be sustained. This directly parallels the threshold $\delta \geq 1/3$: myopic agents with small γ will inevitably learn to defect.

7 Tit-for-Tat

Definition 7.1 (Tit-for-Tat)**.** In a two-player game, the *tit-for-tat* strategy for player i is:

$$s_i^1 = C, \qquad s_i^t(h^t) = a_j^{t-1} \quad \text{for } t \geq 2,$$

where $j \neq i$. That is, start by cooperating, then replicate the opponent's previous action.

Proposition 7.2 (Tit-for-Tat as NE in Repeated PD)**.** *In the infinitely repeated Prisoner's Dilemma (Example 6.1), the tit-for-tat profile is a Nash equilibrium for $\delta \geq 1/3$.*

Proof. Under the tit-for-tat profile, both players cooperate in every period, yielding payoff 3 for each. Consider player i's best deviation.

One-period deviation at time t. If i defects at time t while j cooperates (since j copies i's previous action, which was C), player i receives 4 in period t. In period $t+1$, player j defects (copying i's defection). If i returns to cooperation in period $t+1$, i receives 0 (cooperating against j's defection). In period $t+2$, j cooperates again (copying i's cooperation in $t+1$), restoring mutual cooperation.

The discounted payoff from this deviation path, starting from period t, is:

$$(1-\delta)\big[4 + \delta \cdot 0 + \delta^2 \cdot 3 + \delta^3 \cdot 3 + \cdots\big] = (1-\delta)\Big[4 + \frac{3\delta^2}{1-\delta}\Big] = 4(1-\delta) + 3\delta^2.$$

Cooperation gives payoff 3. The deviation is unprofitable when $3 \geq 4(1-\delta) + 3\delta^2$, i.e., $3\delta^2 - 4\delta + 1 \leq 0$, i.e., $(3\delta - 1)(\delta - 1) \leq 0$. Since $\delta < 1$, this holds iff $\delta \geq 1/3$.

Permanent defection. If i defects forever starting at t, the payoff is $(1-\delta)[4 + \delta \cdot 1 + \delta^2 \cdot 1 + \cdots] = 4(1-\delta) + \delta = 4 - 3\delta$, and $3 \geq 4 - 3\delta$ iff $\delta \geq 1/3$.

More generally, any deviation sequence can be analyzed period by period. The key observation is that each defection yields at most a one-period gain of 1 (getting 4 instead of 3) followed by a one-period loss of at least 3 (getting 0 instead of 3 when the opponent retaliates), and for $\delta \geq 1/3$ the discounted loss exceeds the gain. □

Remark 7.3. Unlike grim trigger, tit-for-tat is *forgiving*: punishment lasts only one period. However, tit-for-tat is generally *not* subgame perfect in the Prisoner's Dilemma. In the subgame following a deviation (where one player defected and the other cooperated), tit-for-tat prescribes alternating retaliation. In such a subgame, a one-shot deviation to cooperation by the "retaliating" player would restore mutual cooperation sooner, which can be profitable. Nevertheless, tit-for-tat's simplicity and robustness have made it influential in evolutionary game theory and MARL [Shoham and Leyton-Brown, 2008].

8 Nash Folk Theorem

The folk theorems characterize the set of equilibrium payoffs in infinitely repeated games. The Nash folk theorem states that any feasible, strictly individually rational payoff vector can be achieved in a Nash equilibrium for patient enough players.

Nash Folk Theorem

Theorem 8.1 (Nash Folk Theorem). *Let G be a finite normal-form game with feasible payoff set $V = \operatorname{conv}\{g(a) : a \in A\}$. For any payoff vector $v \in V$ satisfying $v_i > \underline{v}_i^{\text{mix}}$ for all i (strictly individually rational), there exists $\underline{\delta} \in (0,1)$ such that for all $\delta \in (\underline{\delta}, 1)$, there is a Nash equilibrium of $G^\infty(\delta)$ with payoff vector v.*

Proof. The proof constructs a Nash equilibrium strategy profile that yields payoff v and deters deviations via permanent minimax punishment.

Step 1: Achieving v through action sequences. Since $v \in V = \operatorname{conv}\{g(a) : a \in A\}$, there exist action profiles $a^{(1)}, \ldots, a^{(K)} \in A$ and weights $\lambda_1, \ldots, \lambda_K \geq 0$ with $\sum_{k=1}^K \lambda_k = 1$ such that $v = \sum_{k=1}^K \lambda_k\, g(a^{(k)})$.

Construct a periodic cycle of length L: action profile $a^{(k)}$ is played for $L_k = \lfloor \lambda_k L \rfloor$ periods (adjusting one L_k so that $\sum_k L_k = L$). Define the action sequence $(a^t)_{t=1}^\infty$ by repeating this cycle. The time-average payoff of this sequence is $\bar{v}(L) = \frac{1}{L}\sum_k L_k\, g(a^{(k)})$, which satisfies $\|\bar{v}(L) - v\| \leq C/L$ for a constant C depending only on the game. Fix $\varepsilon > 0$ small enough that $v_i - \varepsilon > \underline{v}_i^{\text{mix}}$ for all i, and choose L large enough that $\|\bar{v}(L) - v\| < \varepsilon/2$.

As $\delta \to 1$, the normalized discounted payoff of any periodic sequence converges to its time-average (since $(1-\delta)\sum_{t=1}^\infty \delta^{t-1} f(t)$ converges to $\frac{1}{L}\sum_{t=1}^L f(t)$ for any L-periodic sequence f). Choose $\underline{\delta}_1 \in (0,1)$ such that for $\delta > \underline{\delta}_1$, the discounted payoff is within $\varepsilon/2$ of the time-average. Then the discounted payoff is within ε of v.

Step 2: Punishment strategies. For each player i, let $\alpha_{-i}^{m,i} \in \Delta(A_{-i})$ be a mixed minimax strategy profile: $\max_{a_i} \mathbb{E}_{\alpha_{-i}^{m,i}}[g_i(a_i, a_{-i})] = \underline{v}_i^{\text{mix}}$.

Step 3: Strategy construction. Define the following strategy for each player j:

- *Cooperation phase:* Play the periodic sequence $(a^t)_{t=1}^\infty$ from Step 1.
- *Punishment phase:* If some player i is the first to deviate from the prescribed sequence at period $\hat{t}$, then for all $t > \hat{t}$, all players $j \neq i$ play according to $\alpha_{-i}^{m,i}$ (punishing i forever). Player i plays any best response.

Step 4: Verifying the Nash equilibrium condition. If player i follows the prescribed strategy, she receives a payoff within ε of v_i (for $\delta > \underline{\delta}_1$), hence at least $v_i - \varepsilon > \underline{v}_i^{\text{mix}}$.

If player i deviates at period $\hat{t}$, she receives at most $M = \max_{a,i} |g_i(a)|$ for finitely many periods (the deviation period), then enters permanent punishment yielding at most $\underline{v}_i^{\text{mix}}$ per period. The deviation payoff is bounded by:

$$u_i^{\text{dev}} \leq (1-\delta)\sum_{\tau=1}^{\hat{t}} \delta^{\tau-1} M + \delta^{\hat{t}}\, \underline{v}_i^{\text{mix}} \leq (1-\delta^{\hat{t}})M + \delta^{\hat{t}}\, \underline{v}_i^{\text{mix}}.$$

The worst case for the deviator is $\hat{t} = 1$: $u_i^{\text{dev}} \leq (1-\delta)M + \delta\, \underline{v}_i^{\text{mix}}$. As $\delta \to 1$, this converges to $\underline{v}_i^{\text{mix}} < v_i - \varepsilon$. Hence there exists $\underline{\delta}_2 \in (0,1)$ such that for $\delta > \underline{\delta}_2$, the deviation payoff is strictly below the equilibrium payoff.

Setting $\underline{\delta} = \max(\underline{\delta}_1, \underline{\delta}_2)$ completes the proof. □

Remark 8.2. The equilibrium constructed above is Nash but *not* necessarily subgame perfect. The punishing players carry out permanent minimax punishment, but they may have no incentive

to do so: the punishment strategy may not be a best response for the punishers in the punishment phase. For example, if minimaxing player i requires player j to play an action that gives j a low stage payoff, j might prefer to deviate from the punishment. The subgame perfect folk theorem addresses this shortcoming.

9 Subgame Perfect Folk Theorem

Subgame Perfect Folk Theorem

Theorem 9.1 (Subgame Perfect Folk Theorem [Fudenberg and Maskin, 1986]). *Let G be a finite n-player game satisfying the full dimensionality condition (Definition 5.5). For any feasible payoff vector $v \in V$ with $v_i > \underline{v}_i^{\text{mix}}$ for all i, there exists $\underline{\delta} \in (0,1)$ such that for all $\delta \in (\underline{\delta}, 1)$, there is a subgame perfect equilibrium of $G^\infty(\delta)$ with payoff vector v.*

Proof. The proof is constructive. We design a strategy profile with multiple phases—cooperation and player-specific punishment—arranged so that every phase is self-enforcing.

Step 1: Choosing reward payoffs. Fix $v \in V$ with $v_i > \underline{v}_i^{\text{mix}}$ for all i. Since V has full dimension (nonempty interior in $\mathbb{R}^n$), for each player i there exists a feasible payoff vector $w^i \in V$ satisfying:

(i) $w_j^i > \underline{v}_j^{\text{mix}}$ for all $j \neq i$ (punishers receive strictly above their minimax values during the punishment of i);

(ii) $w_i^i < v_i$ (player i's payoff during her own punishment phase is below her cooperative payoff).

The full dimensionality condition guarantees such vectors exist: since V has nonempty interior, we can perturb any interior point of V to independently adjust each coordinate. Specifically, let v^0 be an interior point of V with $v_j^0 > \underline{v}_j^{\text{mix}}$ for all j. For each i, define $w^i = v^0 - \eta e_i$, where e_i is the i-th standard basis vector and $\eta > 0$ is small enough that $w^i \in V$, $w_i^i > \underline{v}_i^{\text{mix}}$, and $w_i^i < v_i$. This is possible because v^0 is interior and $v_i > \underline{v}_i^{\text{mix}}$.

For each i, let $m_{-i}^i \in \Delta(A_{-i})$ be a mixed minimax strategy profile against player i, and let m_i^i be any best response by i to m_{-i}^i.

Step 2: Strategy construction (three-phase automaton).

Phase 0 (Cooperation): Play the action sequence yielding payoff v (constructed as in the Nash folk theorem proof, Step 1).

Phase I_i *(Punishment of player i):* Lasts for L periods (to be determined). During this phase, the opponents of i play the minimax strategy m_{-i}^i each period, and player i plays any best response m_i^i (yielding i at most $\underline{v}_i^{\text{mix}}$ per period).

Phase II_i *(Reward after punishing i):* After the L-period punishment of i, play switches to the action sequence yielding payoff w^i forever.

Transitions:

- If player j is the first to deviate from the prescribed actions in Phase 0, transition to Phase I_j.
- If player $j \neq i$ deviates during Phase I_i, transition to Phase I_j (punish the deviating punisher).
- If player i deviates during her own punishment Phase I_i, restart Phase I_i from the beginning.
- If any player j deviates during Phase II_i, transition to Phase I_j.

Step 3: Choosing the punishment length L. Let $M = \max_{a,i} |g_i(a)|$ be a bound on stage-game payoffs. We need L large enough that the punishment phase is a genuine deterrent. The continuation payoff for player i at the start of Phase I_i is:

$$p_i(\delta, L) = (1 - \delta^L)\, \underline{v}_i^{\mathrm{mix}} + \delta^L\, w_i^i.$$

Since $\underline{v}_i^{\mathrm{mix}} \le w_i^i < v_i$, we have $p_i(\delta, L) \le w_i^i < v_i$ for all L and δ. Moreover, as $L \to \infty$ with δ fixed, $p_i(\delta, L) \to \underline{v}_i^{\mathrm{mix}}$. We can choose L large enough that the punishment is sufficiently severe.

Step 4: Verifying sequential rationality.

We apply the one-shot deviation principle: a strategy profile in a discounted repeated game is an SPE if and only if no player has a profitable single-period deviation at any history [Fudenberg and Tirole, 1991].

Phase 0 – Player i considers deviating. Compliance yields payoff (within ε of) v_i. A one-shot deviation yields at most M in the current period, followed by Phase I_i:

$$u_i^{\mathrm{dev}} = (1 - \delta)\, M + \delta\, p_i(\delta, L).$$

As $\delta \to 1$, $(1 - \delta)M \to 0$ and $\delta\, p_i(\delta, L) \to p_i(1, L) = w_i^i < v_i$. Hence for δ close enough to 1, $u_i^{\mathrm{dev}} < v_i$ and deviation is unprofitable.

Phase I_i – Punisher $j \neq i$ considers deviating. Compliance means playing the minimax action for the remaining $L' \le L$ periods of Phase I_i, then entering Phase II_i with payoff $w_j^i > \underline{v}_j^{\mathrm{mix}}$. A one-shot deviation yields at most M in one period, then Phase I_j begins:

$$u_j^{\mathrm{dev}} = (1 - \delta)\, M + \delta\, p_j(\delta, L).$$

The compliance payoff from the continuation of Phase I_i plus Phase II_i converges to w_j^i as $\delta \to 1$ (the finite punishment phase becomes negligible). Since $w_j^i > \underline{v}_j^{\mathrm{mix}} \ge p_j(\delta, L)$ (the latter being a weighted average of $\underline{v}_j^{\mathrm{mix}}$ and w_j^j, both at most close to $\underline{v}_j^{\mathrm{mix}}$ for large L), compliance dominates deviation for δ large enough.

More precisely: player j's compliance continuation payoff from Phase I_i with L' periods remaining is

$$c_j = (1 - \delta^{L'})\, g_j^{\mathrm{I}_i} + \delta^{L'}\, w_j^i,$$

where $g_j^{\mathrm{I}_i}$ is j's per-period payoff during Phase I_i (from playing the minimax strategy against i). As $\delta \to 1$, $c_j \to w_j^i > \underline{v}_j^{\mathrm{mix}}$. The deviation payoff converges to $p_j(1, L)$, which can be made close to $\underline{v}_j^{\mathrm{mix}}$ by choosing L large. Hence compliance is strictly preferred for large δ.

Phase I_i – Player i considers deviating. Deviation restarts Phase I_i, gaining at most M in one period but adding L more punishment periods. Compliance leads to Phase II_i after L' periods. The compliance continuation payoff is $(1 - \delta^{L'})\, \underline{v}_i^{\mathrm{mix}} + \delta^{L'}\, w_i^i$. The deviation payoff is $(1 - \delta)M + \delta\, p_i(\delta, L)$. For large δ, the compliance payoff exceeds the deviation payoff because compliance leads to Phase II_i sooner (after L' rather than L additional periods).

Phase II_i – Any player j considers deviating. This is identical to the Phase 0 analysis with w^i replacing v. Since $w_j^i > \underline{v}_j^{\mathrm{mix}}$ for all j, the argument from Phase 0 applies.

Step 5: Combining. For each of the finitely many deviation scenarios above, there is a threshold $\delta_k \in (0, 1)$ above which deviation is unprofitable. Setting $\underline{\delta} = \max_k \delta_k$ completes the proof. □

Remark 9.2 (Two-Player Games). For two-player games, the full dimensionality condition is not needed: the subgame perfect folk theorem holds for any two-player game [Fudenberg and Maskin, 1986]. The reason is that in two-player games, the only punisher is the single opponent, and minimaxing the deviator is automatically in the punisher's interest (or at least does not require special incentivization) because Phase II_i can reward the punisher without needing to independently adjust a third player's payoff.

Remark 9.3 (Necessity of Full Dimensionality for $n \geq 3$). For three or more players, the full dimensionality condition is essential. Without it, the feasible payoff set lies in a lower-dimensional subspace, and one cannot independently reward punishers while punishing the deviator. Fudenberg and Maskin [1986] provide a counterexample: a three-player game violating the condition where certain feasible, strictly individually rational payoffs cannot be sustained in any SPE for any δ.

10 Connections to Reinforcement Learning

Self-Play and Emergent Cooperation

In multi-agent RL, agents trained via self-play in iterated games effectively search over strategies in a repeated game. The folk theorems predict that:

(i) **Many equilibria exist.** For patient agents (large γ), the set of possible converged outcomes is enormous. Small changes in learning rates, initialization, or exploration schedules can lead to qualitatively different equilibria.

(ii) **Cooperation can emerge.** Even in social dilemmas (e.g., iterated Prisoner's Dilemma), self-play can discover cooperative strategies resembling grim trigger or tit-for-tat, provided the discount factor is large enough.

(iii) **Punishment structures appear naturally.** Trained agents often develop implicit punishment mechanisms: they cooperate when the opponent cooperates and retaliate after defection, mirroring the trigger strategies in the folk theorem proofs.

Opponent Modeling

The folk theorem strategies require each player to condition on the *entire history* of past play. In RL, this corresponds to:

- **Recurrent architectures** (LSTMs, GRUs) that maintain a hidden state summarizing interaction history.
- **Opponent modeling**: explicitly building a model of the opponent's strategy from observed actions, analogous to detecting whether the opponent is "cooperating" or "punishing" in the folk theorem construction.
- **Theory of mind**: higher-order reasoning about what the opponent believes about one's own strategy, corresponding to the recursive reasoning in SPE verification.

The Discount Factor and Cooperation

The threshold $\underline{\delta}$ in the folk theorems has a direct analog in MARL. Empirically, increasing the discount factor γ in multi-agent training makes cooperative outcomes more likely, consistent with the theoretical prediction. Conversely, highly myopic agents (γ near 0) behave as if playing a one-shot game, converging to the stage-game NE. This provides a principled way to influence emergent behavior through hyperparameter selection [Shoham and Leyton-Brown, 2008].

11 Exercises

Exercise 11.1 (Unraveling in a Specific Game)**.** Consider the stage game with payoffs:

	L	R
T	$2,2$	$0,3$
B	$3,0$	$1,1$

(a) Verify that (B, R) is the unique Nash equilibrium.

(b) Apply the unraveling theorem to determine the unique SPE of the 100-fold repetition G^{100}.

(c) Compute the SPE payoff for each player and compare it with the Pareto-optimal outcome (T, L).

(d) Explain intuitively why the backward induction argument fails if the game is repeated infinitely.

Exercise 11.2 (Grim Trigger Threshold)**.** Consider a symmetric Prisoner's Dilemma with general payoffs:

	C	D
C	b, b	$0, b+c$
D	$b+c, 0$	c, c

where $b > c > 0$ (so mutual cooperation is preferred, but defection is individually tempting).

(a) Compute the minimax value for each player.

(b) Derive the minimum discount factor $\underline{\delta}$ for grim trigger to sustain (C, C) as an SPE.

(c) Show that $\underline{\delta} \to 1$ as $c \to b$, and interpret this result.

(d) For $b = 5, c = 2$, compute the numerical threshold.

Exercise 11.3 (Tit-for-Tat is Not SPE)**.** Using the Prisoner's Dilemma of Example 6.1:

(a) Consider the subgame reached after the history (C, D) (player 1 cooperated, player 2 defected in period 1). Describe the play under tit-for-tat from this point.

(b) Compute the payoffs for both players under tit-for-tat in this subgame.

(c) Show that player 2 can profitably deviate in this subgame (for suitable δ), proving that tit-for-tat is not SPE.

(d) Explain the conceptual difference between NE and SPE in this context.

Exercise 11.4 (Feasible Set Computation)**.** Consider the three-action game:

	L	C	R
T	$4,0$	$0,4$	$1,1$
M	$0,0$	$3,3$	$0,0$
B	$1,1$	$0,0$	$2,2$

(a) List all nine pure action payoff vectors.

(b) Sketch the feasible payoff set $V = \text{conv}\{g(a) : a \in A\}$ in $\mathbb{R}^2$.

(c) Compute the mixed-strategy minimax value for each player.

(d) Identify the set of feasible, strictly individually rational payoff vectors.

(e) Does the full dimensionality condition hold? Why or why not?

Exercise 11.5 (Beyond Grim Trigger: Limited Punishment)**.** In the repeated Prisoner's Dilemma (Example 6.1), consider a *K-period punishment* strategy: cooperate; upon any deviation, defect for exactly K periods, then return to cooperation.

(a) Compute the minimum discount factor $\underline{\delta}(K)$ for this strategy to be an SPE, as a function of K.

(b) Show that $\underline{\delta}(K) \to 1/3$ as $K \to \infty$ (recovering the grim trigger threshold).

(c) Compare $\underline{\delta}(1)$ (single-period punishment) with $\underline{\delta}(\infty) = 1/3$. What is the cost of forgiveness?

(d) Discuss the trade-off between punishment length and the required patience level.

Exercise 11.6 (Folk Theorem Application)**.** In the game of Exercise 11.4:

(a) Using the Nash folk theorem, argue that the payoff vector $(3, 3)$ (from the action profile (M, C)) is achievable as a NE payoff for large δ, even if (M, C) is not a stage-game NE.

(b) Construct an explicit grim trigger strategy profile that yields payoff close to $(3, 3)$.

(c) Argue that the payoff $(2, 2)$ is also achievable as an SPE payoff for large δ, using the subgame perfect folk theorem.

(d) Is the payoff $(0.5, 0.5)$ achievable as an equilibrium payoff? Justify your answer using the minimax lower bound.

Summary. This lesson developed: (1) the finitely repeated game model with the unraveling theorem showing that unique stage-game NE leads to unique SPE; (2) the infinitely repeated game model with discounting; (3) feasible and individually rational payoff sets, with the minimax value providing a payoff lower bound in any NE; (4) grim trigger and tit-for-tat as concrete mechanisms for sustaining cooperation; (5) the Nash folk theorem, showing that any feasible, strictly individually rational payoff is achievable in NE for patient players; and (6) the subgame perfect folk theorem of Fudenberg and Maskin [1986], extending this to SPE under the full dimensionality condition. These results are foundational for understanding emergent behavior in multi-agent RL, where repeated interaction, discount factors, and implicit punishment strategies play central roles.

References

Drew Fudenberg and Eric Maskin. The folk theorem in repeated games with discounting or with incomplete information. *Econometrica*, 54(3):533–554, 1986.

Drew Fudenberg and Jean Tirole. *Game Theory*. MIT Press, 1991.

Yoav Shoham and Kevin Leyton-Brown. *Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations*. Cambridge University Press, 2008.